

PAPER_317: Orion M42 Trapezium Wind Ram Pressure Dominance

$\eta_{\text{wind}} = 28.47$ | $t_{\text{erosion}} = 467 \text{ kyr}$ | $a_{\text{wind}} = 5.424 \times 10^{-10} \text{ m/s}^2$

FIRST UQFF HII Region Ram Pressure Dominance Ratio

Session: 91

Module: ORION_UQFF_MODULE.cpp (33rd C++ module)

System: Orion Nebula M42/NGC 1976 — compact HII region, Trapezium OB cluster ionization source

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Abstract

Within the UQFF framework, this paper quantifies the ram pressure dominance of the Trapezium-driven stellar wind over self-gravity in the Orion Nebula. The dimensionless wind-gravity ratio $\eta_{\text{wind}} = P_{\text{ram}}/P_{\text{grav}} = 28.47$ demonstrates that the HII region was born in a wind-dominated (unbound) state. The erosion timescale $t_{\text{erosion}} = 467 \text{ kyr}$ shows that protoplanetary discs (proplyds) currently observed at $\sim 300 \text{ kyr}$ age survive inside a wind-dominated environment. This is the FIRST UQFF computation of an HII region ram pressure dominance ratio.

System Parameters

Parameter	Value	Description
M	$2000 M_{\text{sun}} = 3.978 \times 10^{33} \text{ kg}$	Total nebular mass
r	$1.18 \times 10^{17} \text{ m}$ ($\sim 12.5 \text{ ly}$)	Half-span
ρ_{fluid}	$1 \times 10^{-20} \text{ kg/m}^3$	HII gas density
v_{wind}	$8 \times 10^3 \text{ m/s}$	Ionization front expansion speed
t_{age}	$3 \times 10^5 \text{ yr} = 9.467 \times 10^{12} \text{ s}$	Nebula age
G	$6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	Gravitational constant

Key Equations

Base Gravity

$$g_{\text{base}} = \frac{GM}{r^2} = \frac{6.6743 \times 10^{-11} \times 3.978 \times 10^{33}}{(1.18 \times 10^{17})^2} = 1.907 \times 10^{-11} \text{ m/s}^2$$

Ram Pressure Acceleration (t = 0)

$$a_{\text{wind}}(t = 0) = \frac{v_{\text{wind}}^2}{r} = \frac{(8 \times 10^3)^2}{1.18 \times 10^{17}} = 5.424 \times 10^{-10} \text{ m/s}^2$$

Ram Pressure Acceleration (t = t_age)

$$a_{\text{wind}}(t_{\text{age}}) = \frac{v_{\text{wind}}^2}{r} \left(1 + \frac{t_{\text{age}}}{t_{\text{age}}} \right) = 2 \times 5.424 \times 10^{-10} = 1.085 \times 10^{-9} \text{ m/s}^2$$

Ram Pressure (Pa)

$$P_{\text{ram}} = \rho \cdot v_{\text{wind}}^2 = 10^{-20} \times (8 \times 10^3)^2 = 6.4 \times 10^{-13} \text{ Pa}$$

Gravitational Pressure (Pa)

$$P_{\text{grav}} = \frac{GM\rho}{r} = \frac{6.6743 \times 10^{-11} \times 3.978 \times 10^{33} \times 10^{-20}}{1.18 \times 10^{17}} = 2.248 \times 10^{-14} \text{ Pa}$$

Wind-Gravity Dominance Ratio (PAPER_317 Key Result)

$$\eta_{\text{wind}} = \frac{P_{\text{ram}}}{P_{\text{grav}}} = \frac{a_{\text{wind}}}{g_{\text{base}}} = \frac{5.424 \times 10^{-10}}{1.907 \times 10^{-11}} = \boxed{28.47}$$

Erosion Timescale

$$t_{\text{erosion}} = \frac{r}{v_{\text{wind}}} = \frac{1.18 \times 10^{17}}{8 \times 10^3} = 4.675 \times 10^{13} \text{ s} = \boxed{467 \text{ kyr}}$$

Kinetic-to-Gravitational Energy Ratio

$$\frac{W_{\text{KE}}}{W_{\text{grav}}} \approx \frac{a_{\text{wind}}}{g_{\text{base}}} = 28.47$$

Results Summary

Quantity	Value	Significance
g_base	$1.907 \times 10^{-11} \text{ m/s}^2$	Newtonian self-gravity
a_wind(t=0)	$5.424 \times 10^{-10} \text{ m/s}^2$	Initial ram pressure
$\eta_{\text{wind}}(t=0)$	28.47	Wind » gravity at birth
a_wind(t_age)	$1.085 \times 10^{-9} \text{ m/s}^2$	Ram pressure at 300 kyr
$\eta_{\text{wind}}(t_{\text{age}})$	56.9	Wind dominance doubles
t_erosion	467 kyr	Pröplyd lifetime > t_age
P_ram	$6.4 \times 10^{-13} \text{ Pa}$	Ram pressure
P_grav	$2.248 \times 10^{-14} \text{ Pa}$	Gravitational pressure

Quantity	Value	Significance
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UQFF Physical Interpretation

The Orion Nebula at $t_{\text{age}} = 300$ kyr is still **28.5-57 $\times$ wind-dominated** (η_{wind} ranging from 28.47 at $t=0$ to 56.9 at t_{age}). The erosion timescale $t_{\text{erosion}} = 467$ kyr $> t_{\text{age}} = 300$ kyr confirms that proplyds currently observed by HST survive because they have not yet been fully ablated by the ram pressure — consistent with observational evidence of ~ 150 – 180 proplyds in the Orion Nebula.

UQFF Significance: This is the FIRST UQFF quantification of wind ram pressure dominance over self-gravity in a compact HII region. The wind term $a_{\text{wind}}(t) = v_{\text{wind}}^2/r \times (1+t/t_{\text{age}})$ [registered as WOLFRAM_TERM_ORION_WIND_RAM in ORION_UQFF_MODULE.cpp] dominates the gravitational term by a factor of 28.47 at birth, growing to 56.9 at 300 kyr — placing Orion in the same UQFF “wind-dominant” class as post-AGB bipolar PNe but via HII ionization physics rather than stellar wind shocks.

WOLFRAM_TERM Registration

```
#define WOLFRAM_TERM_ORION_WIND_RAM(val) (val)
// wind = v_wind^2/r * (1+t/t_age) [PAPER_317 ram pressure dominance; eta_wind=28.47]
```

Series first: FIRST UQFF HII region ram pressure dominance ratio. Distinguishes compact OB-driven HII ($\eta_{\text{wind}}=28.47$) from extended GMC HII and from bipolar PN wind shocks ($\eta_{\text{wind}} \sim 7 \times 10^5$, PAPER_311).